

PROJECT 4

POWER BI DASHBOARD — E-COMMERCE SALES & PERFORMANCE

This project is the final stage of the DecodeLabs Data Analytics Internship. Using the cleaned e-commerce dataset from Project 1 (1,200 rows, 13 columns), a two-page interactive Power BI dashboard was built to visualize business performance across sales, products, customers, payment channels, referral sources, and coupon usage. All DAX measures were written from scratch and stored in a dedicated KPIs table.

STEPS TAKEN

1. Loaded Cleaned Data into Power BI Desktop

- Imported the cleaned Excel table (Cleaned Data) directly into Power BI Desktop.
- Created a separate KPIs table to house all DAX measures.

2. Created a Calculated Column

- Added a Coupon Type column to the Cleaned Data table to classify each order:

Coupon Type = IF('Cleaned Data'[Coupon Code] = "NO COUPON", "No Coupon", "With Coupon")

- This column was used to segment coupon vs non-coupon orders across visuals.

3. Wrote DAX Measures (13 Total)

Core KPIs:

- Total Revenue = SUM('Cleaned Data'[Total Price])
- Average Order Value = DIVIDE([Total Revenue], [Total Orders])
- Total Orders = DISTINCTCOUNT('Cleaned Data'[Order ID])
- Total Customers = DISTINCTCOUNT('Cleaned Data'[Customer ID])
- Total Products Sold = SUM('Cleaned Data'[Quantity])

Order Status Rates:

- Cancellation Rate = DIVIDE(CALCULATE(COUNTROWS('Cleaned Data'), 'Cleaned Data'[Order Status] = "Cancelled"), COUNTROWS('Cleaned Data')) * 100
- Delivery Rate = DIVIDE(CALCULATE(COUNTROWS('Cleaned Data'), 'Cleaned Data'[Order Status] = "Delivered"), COUNTROWS('Cleaned Data')) * 100

Coupon Breakdown:

- With Coupon = `CALCULATE(COUNTROWS('Cleaned Data'), FILTER('Cleaned Data', 'Cleaned Data'[Coupon Code] <> "No Coupon"))`
- Without Coupon = `CALCULATE(COUNTROWS('Cleaned Data'), FILTER('Cleaned Data', 'Cleaned Data'[Coupon Code] = "No Coupon"))`
- FREESHIP = `CALCULATE(COUNTROWS('Cleaned Data'), FILTER('Cleaned Data', 'Cleaned Data'[Coupon Code] = "FREESHIP"))`
- SAVE10 = `CALCULATE(COUNTROWS('Cleaned Data'), FILTER('Cleaned Data', 'Cleaned Data'[Coupon Code] = "SAVE10"))`
- WINTER15 = `CALCULATE(COUNTROWS('Cleaned Data'), FILTER('Cleaned Data', 'Cleaned Data'[Coupon Code] = "WINTER15"))`

4. Page 1: Executive Overview

- 5 KPI cards: Total Revenue, Avg Order Value, Total Orders, Total Customers, Total Products Sold.
- Area line chart — Monthly Revenue Trend with a year slicer (2023, 2024, 2025).
- Horizontal bar chart — Top Products by Revenue.
- Table — Top 10 Customers by Revenue (Customer ID and Items in Cart).
- Table — Order Record showing full transaction log.
- Slicers: Payment Method, Referral Source.

5. Page 2: Order Analysis

- 5 KPI cards (same set as Page 1).
- Bar chart — Referral Source Performance.
- Bar chart — Sales by Payment Channel.
- Donut chart — Coupon Type breakdown (With Coupon vs No Coupon).
- Bar chart — Order Status Distribution.
- Text box — 5-Point Key Insights summary.
- Year slicer: 2023, 2024, 2025.

KEY FINDINGS

1. Overall Performance (All Years, Unfiltered)

- Total Revenue: \$1.26M
- Average Order Value: \$1.05K
- Total Orders: 1,200

- Total Customers: 1,189
- Total Products Sold: 3,535

2. Product Performance

- Chair and Printer are tied at the top with \$196K each.
- Followed by Laptop (\$192K), Tablet (\$187K), Monitor (\$176K), Desk (\$167K), Phone (\$152K).
- Revenue is relatively balanced across categories with no single product dominating.

3. Revenue Trend

- Revenue peaks in June (~\$170K) across the combined dataset.
- A sharp drop occurs in July–August before a modest recovery.
- The mid-year decline is the most significant planning risk for the business.

4. Order Status

- Cancelled: 250 | Returned: 247 | Pending: 237 | Shipped: 235 | Delivered: 231.
- Cancelled and returned orders (497 combined) outnumber delivered orders (231).
- Fulfilment failure is the highest-impact operational issue in the dataset.

5. Coupon Usage

- 891 orders (74.25%) were placed using a coupon code.
- Only 309 orders were placed at full price.
- FREESHIP: 313 orders | WINTER15: 292 | SAVE10: 286.
- High coupon dependency suggests customers are being conditioned to wait for discounts.

6. Channel Performance

- Referral sources range from \$226K (Referral) to \$275K (Instagram).
- Payment methods range from \$232K (Debit Card) to \$263K (Credit Card)
- No single channel significantly outperforms the others.